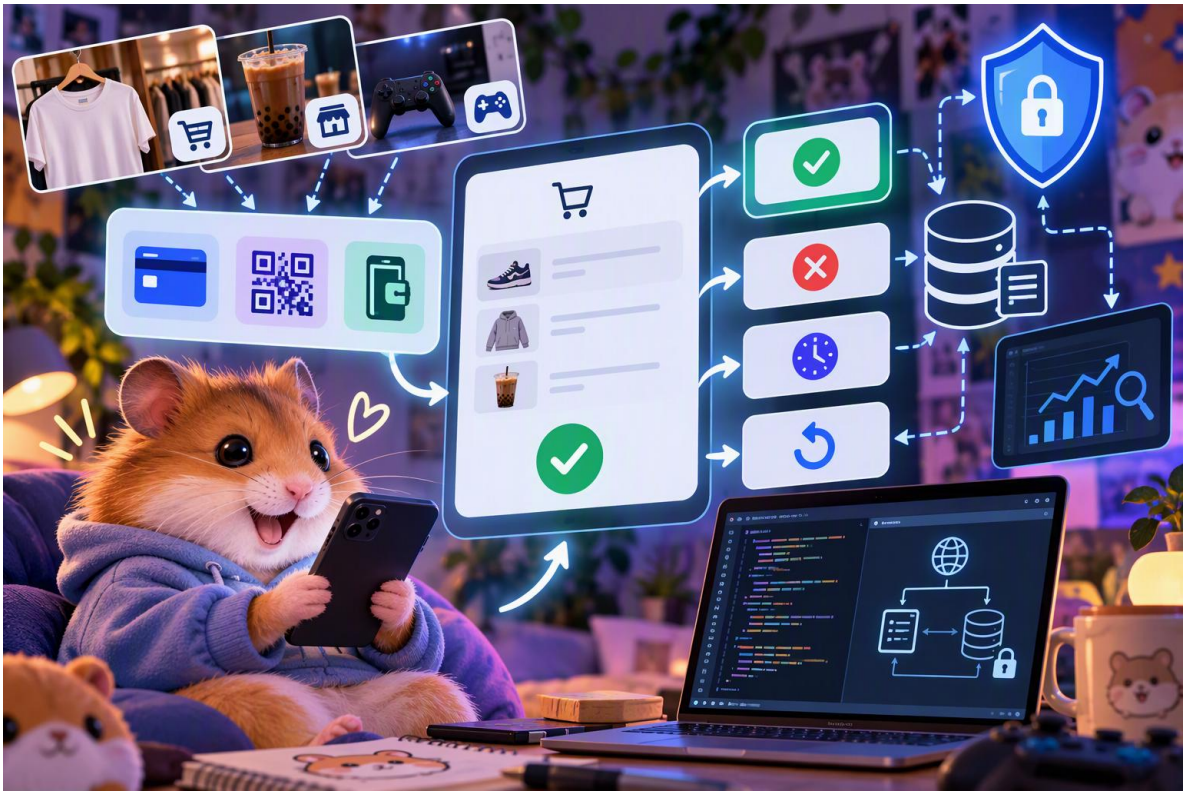
C372 Payment Technologies

Final Assessment Graded Assignment

Digital payments are an essential part of how individuals and businesses purchase goods, access services and manage transactions. From online card payments to QR-based transfers such as PayNow, these technologies support convenient and efficient transactions across a wide range of business contexts.



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However, a successful digital payment solution involves more than displaying a payment button. An application must connect the payment to a business transaction, determine the correct amount, communicate with a payment provider, handle different payment outcomes and maintain accurate records. It must also protect access to sensitive information, support transaction monitoring and manage refunds or cancellations appropriately.

Overview

In this assessment, you will apply these concepts by developing an original payment-enabled web application based on a business idea of your choice. Your goal is to demonstrate a complete and traceable payment workflow supported by appropriate application architecture, database design and security controls.

All payment activities must use test-mode or lecturer-approved simulated environments. You will not process real payments or collect real payment-card details. The emphasis is on implementing, testing and explaining how digital payments support a coherent business solution.

Assessment

In this module, the Final Assessment (Graded Assignment) contributes 55% to the overall module grade. The breakdown of the assessment criteria and their respective weightings is as follows:

Criteria	Description	Weight (%)
Methods and Procedures	How did you approach and develop it?	10%
Quality of Final Deliverables	How well does the completed solution work?	10%
Analysis and Discussion of Knowledge and Skills	How well can you analyse, explain and justify what you built and the decisions you made?	20%
Communication Skills	How effectively can you communicate that understanding?	15%
Total		55%

Project Details

1. Your Business Idea

You may choose any suitable business idea. Your idea must meet all the following conditions:

- Payment is central to the solution, not an optional page added at the end.
- The application addresses a clear user or business need.
- There is at least one identifiable product, service, booking, subscription, membership, contribution or other payable item.
- The business context allows you to identify relevant regulatory and industry requirements, such as PCI DSS, AML/KYC requirements or applicable MAS guidelines.
- The scope is achievable within the module timeframe.
- The scenario allows you to demonstrate all mandatory payment requirements in this brief.
- **Cryptocurrency and blockchain-based payment solutions must not be used for this assignment, as these topics are covered in the C373 module and its related assignment.**

2. Project Application Requirements

All requirements below must be adapted meaningfully to your chosen business context.

a) Application structure and baseline functions

- Organise the application using Model-View-Controller (MVC) principles, with clear routes, controllers, models and views. The application should use the technologies taught in class, including JavaScript, Node.js, Express and EJS.
- Provide the baseline business functions needed for users to reach and complete the payment workflow.
- Prepare wireframes or interface sketches for the key user screens and payment workflow before implementation.
- Use a suitable database design to support users, payable items, business records and payment records.

b) User accounts and access

- Provide user accounts and authentication appropriate to the scenario.
- Ensure users can access only the functions and records appropriate to their role or account.

c) Payable item or service selection

- Allow a user to select, book, subscribe to, contribute towards or purchase at least one payable item or service.
- Display sufficient information for the user to understand what is being paid for.
- Calculate or retrieve the trusted payment amount on the server side rather than accepting an amount supplied only by the browser.

d) Checkout and payment initiation

- Provide a coherent checkout or payment-initiation workflow.
- Show an order, booking or payment summary before payment is initiated.
- Create a traceable business record and payment record for the transaction.

e) Payment-provider integration

- Integrate a suitable payment service provider using its sandbox, test or other non-production environment. Obtain prior approval from your lecturer if you intend to use an alternative payment approach that does not provide such an environment.
- Handle the request to the payment provider and the resulting payment outcome.
- Do not store or process real card details in your application.

f) Payment status handling

- Support and demonstrate APPROVED, PENDING and DECLINED payment outcomes.
- Update the related business record appropriately when the payment status changes.
- Provide clear feedback to the user for each outcome.

g) Persistent payment records and transaction history
- Store payment records persistently in the database.
- Store a suitable transaction or provider reference, amount, payment method, status and date/time.
- Allow users to view their transaction history and relevant transaction details.

h) Security control
- Implement and explain at least one relevant payment or application security control.
- Examples may include secure configuration, access control, server-side validation, password protection or appropriate secret management.
- The selected control must be applied meaningfully, not only described.

i) Fraud detection or transaction monitoring
- Implement at least one fraud-detection or transaction-monitoring rule.
- The rule must identify and record a relevant risk signal, unusual event or suspicious transaction pattern.
- Provide evidence that the rule can be triggered and reviewed.

j) Payment alternative
- Include a QR-based or PayNow-inspired payment alternative or provide an approved architecture comparison where implementation is not required.
- Explain how this alternative differs from the card-provider payment flow used in the application.

k) Refund or reversal
- Provide an appropriate refund, cancellation or reversal workflow.
- Update the transaction and related business status consistently.
- Maintain a traceable record of the post-payment action.

l) Payment reporting
- Provide a payment-focused history, summary or dashboard that supports the chosen scenario.
- Use meaningful transaction information such as status, value, payment method, refund or monitoring information.

Version Control and Deployment

Version control helps you stay organized, track progress, and collaborate effectively without losing work.

- Create a GitHub repository for your project.
- Commit and push your code regularly to show progress.
- Deploy your project on a hosting platform (e.g. Render).
- Ensure the deployed site is publicly accessible.
- Include both the repository link and the live deployment link in your submission.

Regular commits and a working deployment provide evidence of consistent progress and sound technical practice throughout the project.

Project Rules & Guidelines

a) To ensure fairness, authenticity, and technical rigor, students must adhere to the following rules:
   - Code must be original and based on concepts taught in class.
   - Be creative but keep the project achievable within the timeframe.
   - Do not use pre-built or full website templates; marks will be deducted if found.
   - Hardcoded features or data will not earn marks.

b) Use of AI Tools
   Students are permitted to use AI tools (e.g., ChatGPT, GitHub Copilot) to support their work for this assessment. However, any use of AI-generated content must be **clearly acknowledged during the project interview**.

   Students must indicate:

   - The AI tool(s) used
   - The input prompts provided
   - Which parts of the code or content were generated or assisted by AI
   - How the generated responses were adapted or integrated into the work

   Failure to provide acknowledgement of the use of the AI tool(s) is tantamount to plagiarism and should be dealt with in accordance to RP's Rules and Regulations pertaining to academic integrity.

c) Submission Instructions

   a. Milestone 1 - Week 6, submit in **SA3 system**
   b. Milestone 2 - Week 8, submit in **SA3 system**
   c. Final submission, submit in **SA3 system** by **Friday, 29 January 2027, 11:59 PM (2359H)**

   Before you submit, check that you have compressed all required files into a single ZIP file.

   Your ZIP file should follow the naming convention:
   <student_class_code>_<student_name>_<student_id>_FA_GA.zip
   E.g. C372-003_bobochan_2512345_FA_GA.zip

d) Final Deliverables

   a. Short, recorded demonstration video (~7 minutes maximum) explaining the application and showing the key payment flows and relevant post-payment actions.
   b. Complete project source files, including JavaScript, EJS, CSS, images, configuration files and required libraries or dependency definitions.

c. Database SQL scripts required to create and initialise the application database.
d. Graded Assignment Project Report, including key wireframes, technical explanations, supporting evidence, project reflection, the Individual AI Interaction Journal and the Individual Declaration of Compliance.

e) Make sure to back up your project files to avoid any loss of data or project work.

f) Avoid submitting your work at the last minute. Submit early—you can always make changes and resubmit before the deadline.

Penalty for late submissions:

Time after submission deadline, t	0 < t ≤ 24 hours	24 < t ≤ 48 hours	48 < t ≤ 72 hours	t > 72 hours
Downgrading applied to submitted work	10% of the maximum score for the submitted work	20% of the maximum score for the submitted work	30% of the maximum score for the submitted work	100% of the maximum score for the submitted work, i.e., award zero marks

Progressive check-ins

Stage	Expected progress	Indicative timing
Idea and structure	Business need, payable item, baseline features and initial MVC structure	Week 2
Specification and architecture	Requirements, business rules, acceptance criteria and payment architecture	Week 3
Data and interface design	ERD, data model, routes, MVC responsibilities and checkout design	Week 4
Core workflow	Core data and preliminary payment flow	Week 5
Provider integration	Payment integration	Week 6
Security and monitoring	Security control and fraud/monitoring rule	Week 7-8
Payment alternative	QR/PayNow-inspired alternative or approved comparison	Week 9
Completion	Refund, transaction history, final integration and verification	Week 10
Final submission and demonstration	Complete application and technical explanation	Week 11
Interview		Week 12